

Usman Asad

Senior Python & AI Engineer · Lead Software Engineer

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PROFESSIONAL SUMMARY

Senior Python & AI Engineer with **4+ years** designing and delivering production backend and AI systems across enterprise, security, fintech, and SaaS environments. Currently **Lead Developer at ZeusLock.ai** — an AI-native **Data Loss Prevention (DLP)** platform for European enterprise and sovereign clients — where I own a **Python / FastAPI** backend and an **LLM-powered detection engine** running on self-hosted GPU infrastructure. Co-Founder of **Digifyr**, a software studio that has shipped platforms like Ataya, NafaPay, and CargoControl to **20,000+ users**. Deep expertise in **Python, FastAPI, PostgreSQL, SQLAlchemy / Alembic, Celery, Keycloak (OIDC), LLM integration (LangChain, LangGraph, llama.cpp), Docker, Kubernetes, and GitOps CI/CD** — complemented by Next.js / React web and Flutter mobile development.

PROFESSIONAL EXPERIENCE

Lead Developer | ZeusLock.ai

Nov 2025 – Present | Remote / France

AI-native Data Loss Prevention (DLP) platform that stops sensitive-data leaks into AI tools (ChatGPT, Claude, Copilot, MCP); shipped as SaaS and on-premise to European enterprise, defense, and sovereign clients (zeuslock.ai).

- **Lead architecture and delivery** of the full platform: a multi-tenant **Python / FastAPI** backend (21 API routers), **Celery** workers, **PostgreSQL 16** with **Alembic** migrations, **Keycloak (OIDC)** authentication, and Redis / Valkey caching.
- **Built the AI detection engine** — a hybrid pipeline of regex rules, **policy-aware LLM classification**, and an **LLM jailbreak classifier** analyzing prompts and files (PDF, DOCX, XLSX, images) in real time.
- **Self-host the LLM stack** (Gemma via **llama.cpp**, NVIDIA RTX GPU) behind an OpenAI-compatible API — fully sovereign, EU-hosted inference with no third-party AI providers; added **multimodal image / PDF detection** and an **OCR-first pipeline measured 3.3× faster** than vision-model inference.
- **Cut LLM latency ~1.9×** with a hybrid reasoning policy (thinking on/off per request type with automatic retry-escalation), validated against a **602-case detection benchmark** that gates every model and prompt change.
- Engineered **multi-tenant isolation with PostgreSQL Row-Level Security** across 15 org-scoped tables, encrypted API keys, and audit logging; built **format-preserving anonymization** and **SIEM integrations** (Splunk, Sentinel, Elastic / OCSF).
- Own the infrastructure: **Kubernetes (k3s)** on an OVH GPU node, a universal **Helm** chart, **ArgoCD GitOps** deploys, Traefik + cert-manager TLS, and GitHub Actions CI/CD with image signing, SBOM attestation, and vulnerability scanning.
- Ship **SaaS and an air-gap-capable on-premise edition from one codebase**, with Ed25519 license verification and a private Harbor registry; enforced at build and runtime, validated on customer bare-metal installs.
- Also deliver the **Next.js** customer and superadmin dashboards (CMS), Manifest v3 **browser extension**, and Electron **desktop agent** (MITM proxy, **MCP monitoring**, Shadow-AI discovery) enforcing backend decisions at the endpoint.

Co-Founder & Technical Lead | Digifyr

Jul 2022 – Present | Islamabad, Pakistan

Software studio I co-founded delivering SaaS, fintech, and mobile products for clients in finance, logistics, and tourism; lead a small team of engineering consultants on concurrent engagements (digifyr.com).

- Run technical direction across the studio — architecture, code review standards, release cadence — and recruit and mentor engineering consultants.
- **Ataya** — Leisure booking and matching platform: **Python / FastAPI backend + PostgreSQL**, Next.js web, Flutter apps; Stripe / Paytech / Wave / Orange Money payments; **10K+ downloads** and a **40% increase** in bookings; on GCP. ataya.app
- **NafaPay** — Buy-now-pay-later (BNPL) fintech platform for Senegal: REST API backend + **PostgreSQL**, Flutter apps, web marketplace; interest-free installments and Wave / Orange Money payments; **2,000+ downloads**. nafapay.shop
- **CargoControl** — Real-time cargo tracking SaaS (REST APIs + MySQL + Flutter + Next.js); role-based portals, Google Maps live vessel/truck tracking, and FCM alerts; deployed on GCP; improved operational insights by 20%. cargocontrol.io
- **MamanTontine** — Shipped a production fintech platform digitalizing tontines (rotating group-savings circles) for West African markets — backend APIs, React web app, and Flutter mobile. mamantontine.com

Machine Monitoring System (MMS)*Live demo: mms-aim.netlify.app*

- Designed an end-to-end, scalable **Python backend on FastAPI + PostgreSQL** using a microservices architecture and Clean Architecture principles; delivered iteratively via Agile / Scrum.
- Implemented **real-time machine data streaming** over WebSockets with **Redis** and **RabbitMQ** for high-throughput message brokering between the backend and Flutter clients.
- Built a **predictive maintenance** module with an automated multi-channel alert system (WhatsApp, SMS, Email) using **Celery** for scheduled background job processing.
- Secured APIs with **Keycloak (JWT / RBAC)** for role-based access across 1,000+ users; **optimized API performance from 20s → 4s** by refactoring queries and redesigning the schema.
- Containerized services with **Docker**; deployed via **Azure App Service & ACR** with CI/CD; shipped real-time analytics dashboards, reports, and billing — reducing manual reporting effort by **40%**.

Assistant Manager – Software Engineer | NESCOM

Jul 2023 – May 2025 | Islamabad, Pakistan

Mission Planning Software

- Developed an intelligent mission planning system under a structured **Waterfall SDLC**, with **automated path planning** built on classical **genetic algorithms** and DLL-based interoperability with other mission-critical systems; optimized the planning engine for a **33% efficiency gain** and lower mission-generation latency.
- Integrated **ArcGIS** for real-time geospatial visualization with multi-map loading and advanced data overlays, improving situational awareness and decision-making.
- Built a high-performance **C++ navigation utility** implementing **TERCOM (terrain contour matching)** for Windows and Linux mission planning systems, improving real-time execution speed and accuracy.

PROJECTS

FaceRT — Real-Time Face Recognition & Tracking (Final Year Project) · 2022 – 2023

- Built a facial recognition system achieving **98.6% accuracy** for **real-time multi-face detection**, tracking, and entry logging — **RetinaFace + ArcFace** models (TensorFlow) served through a **Python / Flask REST API** with MongoDB, plus React web and Flutter mobile clients. github.com/UsmanAsad87/FaceRecogFYPModel

Deep Learning Stock Price Prediction — Published Research · 2023

- Comparative study of ML models for S&P 500 stock price prediction; the proposed **Single LSTM Engine** outperformed alternatives with **RMSE < 3%**. Stack: Python, TensorFlow / Keras, pandas. bit.ly/40sLfvl

EDUCATION

BSc Computer Science | PIEAS

Sep 2019 – Jun 2023 | Islamabad, Pakistan

Final grade: **3.91 / 4.00** · EQF Level 6 · Fully Funded Fellowship (NESCOM, 2020–2023)**TECHNICAL SKILLS**

Languages: Python, SQL, JavaScript, TypeScript, Dart, C#, C / C++**Python Backend:** FastAPI, SQLAlchemy, Alembic, Pydantic, Celery, asyncio, RESTful APIs, microservices, Clean Architecture, WebSockets, OpenAPI / Swagger, RabbitMQ, Redis / Valkey, Keycloak (OIDC / JWT)**AI & ML:** LLM integration (Llama.cpp, OpenAI-compatible APIs), LangChain, LangGraph, LLM fine-tuning (Unsloth), prompt engineering, jailbreak / guardrail detection, vision + OCR pipelines, TensorFlow / Keras, pandas, NumPy, scikit-learn**Frontend & Mobile:** Next.js, React.js, Tailwind CSS, Flutter, responsive UI/UX**Data:** PostgreSQL (Row-Level Security, multi-tenancy), MySQL, MSSQL, MongoDB, Firebase**Cloud & DevOps:** Docker, Kubernetes (k3s), Helm, ArgoCD (GitOps), Traefik, nginx, GitHub Actions CI/CD, image signing & scanning, Harbor / GHCR, AWS (EC2, S3, RDS), Azure, GCP, OVH**Security & Payments:** OIDC / OAuth2, JWT, RBAC, SIEM (Splunk, Sentinel, Elastic), GDPR, Stripe, Wave, Orange Money, Paytech**CERTIFICATIONS & AWARDS**

Awards: Hafeez Qureshi NESCOM Excellence Award (Jun 2024) · Fully Funded NESCOM Fellowship at PIEAS (2020–2023) · Co-Lead Machine Learning, Google Developer Student Club (2021–2022) · Treasurer, PIEAS AI Society (2022–2023)